

Test cases and results

26 / 26

tests passing

11

integration tests

15

unit-test executions

0

failures on correct code

ID	Integration	Input	Expected	Actual	Status
SIT-01	Customer → Order	Valid login and order	Order created	"Order confirmation sent to John"	Pass
SIT-02	Order → Product	Product ID P100	Price 800.00 retrieved	800.00	Pass
SIT-03	Order → Inventory	Quantity = 2	Stock 10 → 8	8	Pass
SIT-04	Order → Payment	Amount = 1600	Payment successful	true	Pass
SIT-05	Payment → Order	Successful payment	Proceeds to confirmation	Confirmation returned	Pass
SIT-06	Order → Notification	Successful order	Confirmation sent	"Order confirmation sent to John"	Pass
SIT-07	Order → Inventory	Quantity = 15	Order rejected	"Insufficient stock", stock still 10	Pass
SIT-08	Order → Product	Invalid ID (P999)	Order rejected	"Product not found", stock still 10	Pass
SIT-09	Customer → Order	Invalid login	Order not created	"Login failed", stock still 10	Pass
SIT-10	Payment → Order	Invalid amount (qty 0 → 0.00)	Order fails	"Payment failed", stock still 10	Pass

Table 1: System integration test cases and results.

```
$ mvn test    (correct code)

[INFO] Running shop.ModuleUnitTests
[INFO] Tests run: 15, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.111 s -- in shop.ModuleUnitTests
[INFO] Running shop.ShoppingSystemIntegrationTest
[INFO] Tests run: 11, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.026 s -- in shop.ShoppingSystemIntegrationTest
[INFO] Results:
[INFO] Tests run: 26, Failures: 0, Errors: 0, Skipped: 0
[INFO] BUILD SUCCESS
```

Figure 1: All 26 tests pass on the correct code (BUILD SUCCESS).

Main application output

```
$ java -cp target/classes shop.ShoppingSystem

Customer logged in successfully.
Order confirmation sent to John
Remaining stock: 8
```

Figure 2: Console output of ShoppingSystem.